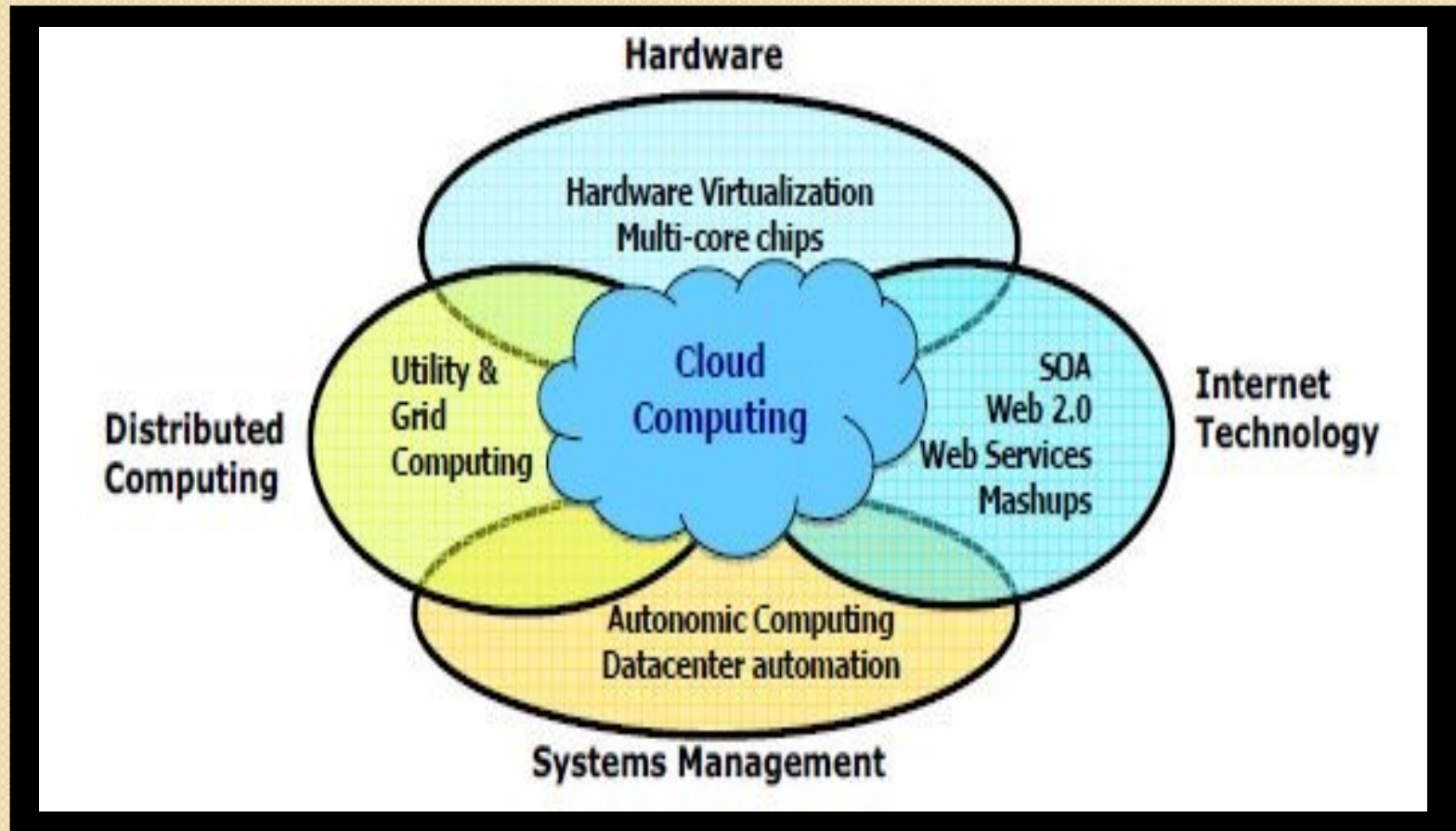


Distributed and Cloud Computing

K. Hwang, G. Fox and J. Dongarra

Enabling Technologies
and Distributed System Models

Data Deluge (Huge Data) Enabling New Challenges



(Courtesy of Judy Qiu, Indiana University, 2011)

So Many Things....

- **Parallel Computing**
- **Distributed Computing**
- **P2P Computing**
- **Cluster Computing**
- **Grid Computing**
- **Cloud Computing**
- **Internet of Things (IoT)**

How to Distinguish...

- **Hardware Architecture**
 - **Uniprocessor, Multiprocessor, Multicomputer, Pipelined, Superscaler, VLIW...**
- **OS Platforms**
 - **UNIX, Mac, Windows, Features**
- **Processing Algorithms**
 - **Centralized, Distributed, Static, Dynamic, Batch**
- **Communication Protocols**
 - **TCP, IP, FTP, OSI**
- **Service Models Applied**

Essential Issues....

- **Scalability**
- **Performance**
- **Availability**
- **Security**
- **Reliability**
- **Energy Efficiency**
- -----

Age of Internet Computing

■ LINPAC Benchmark

“It was designed to help users estimate the time required by their systems to solve a problem using the LINPACK package, by extrapolating the performance results obtained by 23 different computers solving a matrix problem of size 100. This size was chosen due to memory limitations at that time. 10000 floating-point entries from -1 to 1 are randomly generated to fill in a general, dense matrix.”

For each computer system, the following quantities are reported:

- Rmax: the performance in Gflop/s for the largest problem run on a machine .
- Nmax: the size of the largest problem run on a machine .
- N1/2: the size where half the Rmax execution rate is achieved.
- Rpeak: the theoretical peak performance Gflop/s for the machine.

HPC v/s HTC

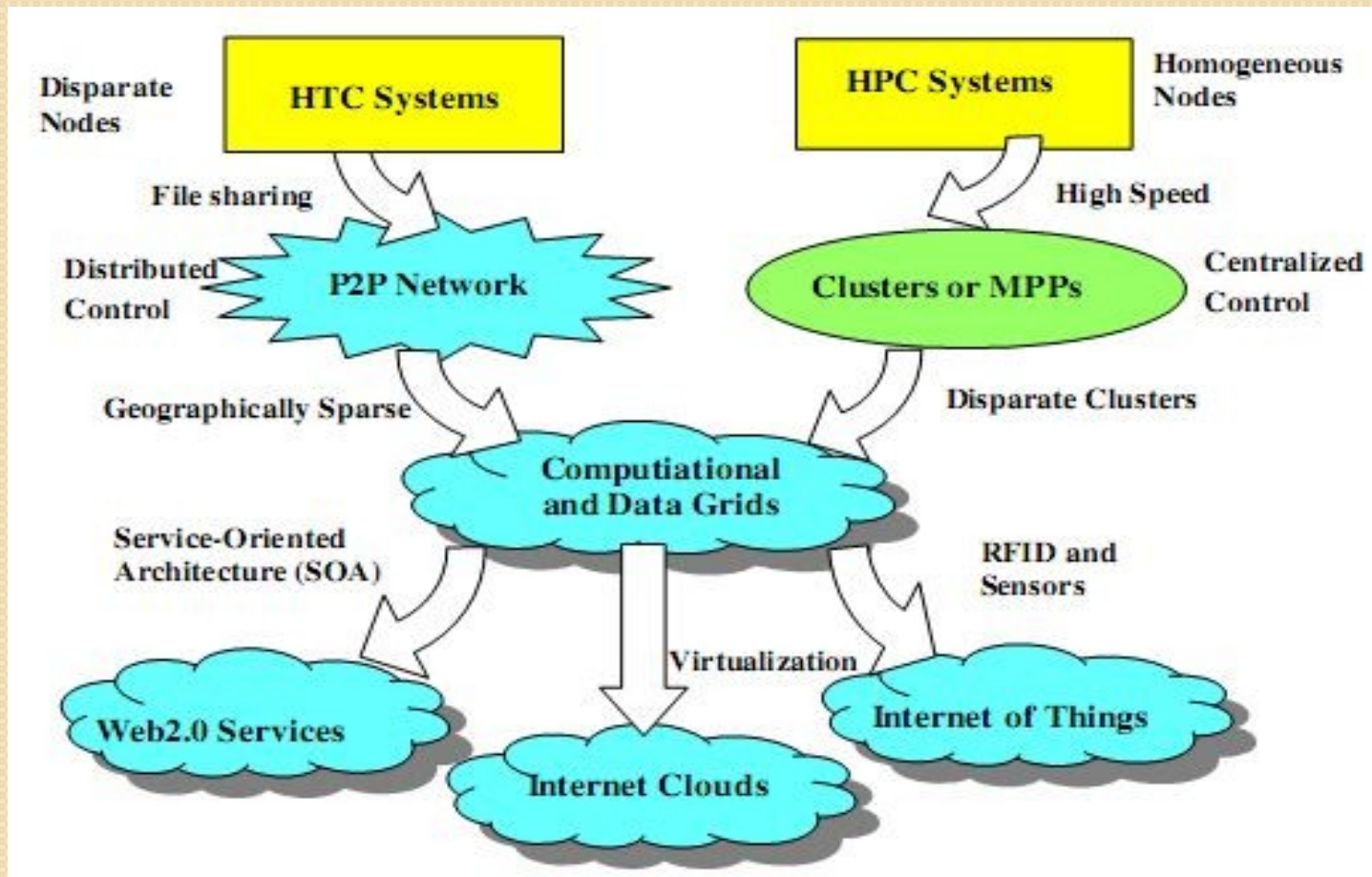
■ HPC

- Raw Speed Performance May be in Gflops
- More focused on increasing the speed of processing

■ HTC

- High Flux Computing
- Use in Internet Searches and Web Services
- Performance goal of high throughput i.e. number of tasks executed per unit time

Clouds and Internet of Things



HPC: High-Performance Computing

HTC: High-Throughput Computing

P2P: Peer to Peer

MPP: Massively Parallel Processors

Source: K. Hwang, G. Fox, and J. Dongarra,
Distributed and Cloud Computing,
Morgan Kaufmann, 2012.

Degrees of Parallelism

- **Bit Level Parallelism**
 - **Bit Serial Processing to Word Level Processing**
 - **4 bit Microprocessors to 8 bit**
- **Instruction Level Parallelism**
 - **Multiple Instructions Executed Simultaneously**
 - **Includes multiple issue super scalar architecture**
- **VLIW**
- **Task Level Parallelism**
- **Data Level Parallelism**
 - **SIMD Machines**
 - **Chip Multiprocessors**
 - **Job Level Parallelism**

From Desktop/HPC/Grids to Internet Clouds in 30 Years

- HPC moving from centralized supercomputers to geographically distributed desktops, desksides, clusters, and grids to clouds over last 30 years
- HTC moving from P2P to more cooperating grid environment
- R/D efforts on HPC, clusters, Grids, P2P, and virtual machines has laid the foundation of cloud computing that has been greatly advocated since 2007
- Location of computing infrastructure in areas with lower costs in hardware, software, datasets, space, and power requirements – moving from desktop computing to datacenter-based clouds

The Trend Towards Utility Computing

- Focuses on the business model in which customers receive computing resources from a paid service provider
- All Grid/Cloud platforms comes under Utility Computing
- Cloud a broader concept than Utility Computing
 - Distributed cloud applications run on any available server in EDGE networks
 - For e.g. users demand new network efficient processors, scalable memory and storage space which is possible with Cloud

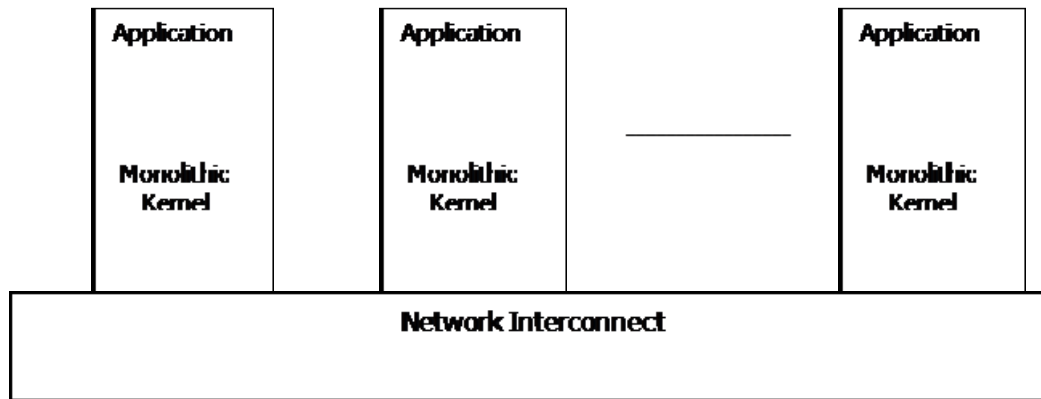
Distributed systems

- Collection of autonomous computers, connected through a network and distribution software called middleware which enables computers to coordinate their activities and to share system resources.
- Characteristics:
 - The users perceive the system as a single, integrated computing facility.
 - The components are autonomous.
 - Scheduling and other resource management and security policies are implemented by each system.
 - There are multiple points of control and multiple points of failure.
 - The resources may not be accessible at all times.
 - Can be scaled by adding additional resources.
 - Can be designed to maintain availability even at low levels of hardware/software/network reliability.

Table 1.2 Classification of Distributed Parallel Computing Systems

Functionality, Applications	Multicomputer Clusters [27, 33]	Peer-to-Peer Networks [40]	Data/Computational Grids [6, 42]	Cloud Platforms [1, 9, 12, 17, 29]
Architecture, Network Connectivity and Size	Network of compute nodes interconnected by SAN, LAN, or WAN, hierarchically	Flexible network of client machines logically connected by an overlay network	Heterogeneous clusters interconnected by high-speed network links over selected resource sites.	Virtualized cluster of servers over datacenters via service-level agreement
Control and Resources Management	Homogeneous nodes with distributed control, running Unix or Linux	Autonomous client nodes, free in and out, with distributed self-organization	Centralized control, server oriented with authenticated security, and static resources	Dynamic resource provisioning of servers, storage, and networks over massive datasets
Applications and network-centric services	High-performance computing, search engines, and web services, etc.	Most appealing to business file sharing, content delivery, and social networking	Distributed super-computing, global problem solving, and datacenter services	Upgraded web search, utility computing, and outsourced computing services
Representative Operational Systems	Google search engine, SunBlade, IBM Road Runner, Cray XT4, etc.	Gnutella, eMule, BitTorrent, Napster, KaZaA, Skype, JXTA, and .NET	TeraGrid, GriPhyN, UK EGEE, D-Grid, ChinaGrid, etc.	Google App Engine, IBM Bluecloud, Amazon Web Service(AWS), and Microsoft Azure,

Multiprocessor and Multicomputer



■ Multi Processor Systems

- Tightly Coupled
- Processors on the Same Board
- Share the Same Clock, Memory and Resources
- Uniform Memory Access

■ Multi Computer Systems

- Loosely Coupled
- Autonomous Computers with its own OS
- Non Uniform Memory Access
- Provides Strong Computational Power with Certain Degree of Resource Independence

A Typical Cluster Architecture

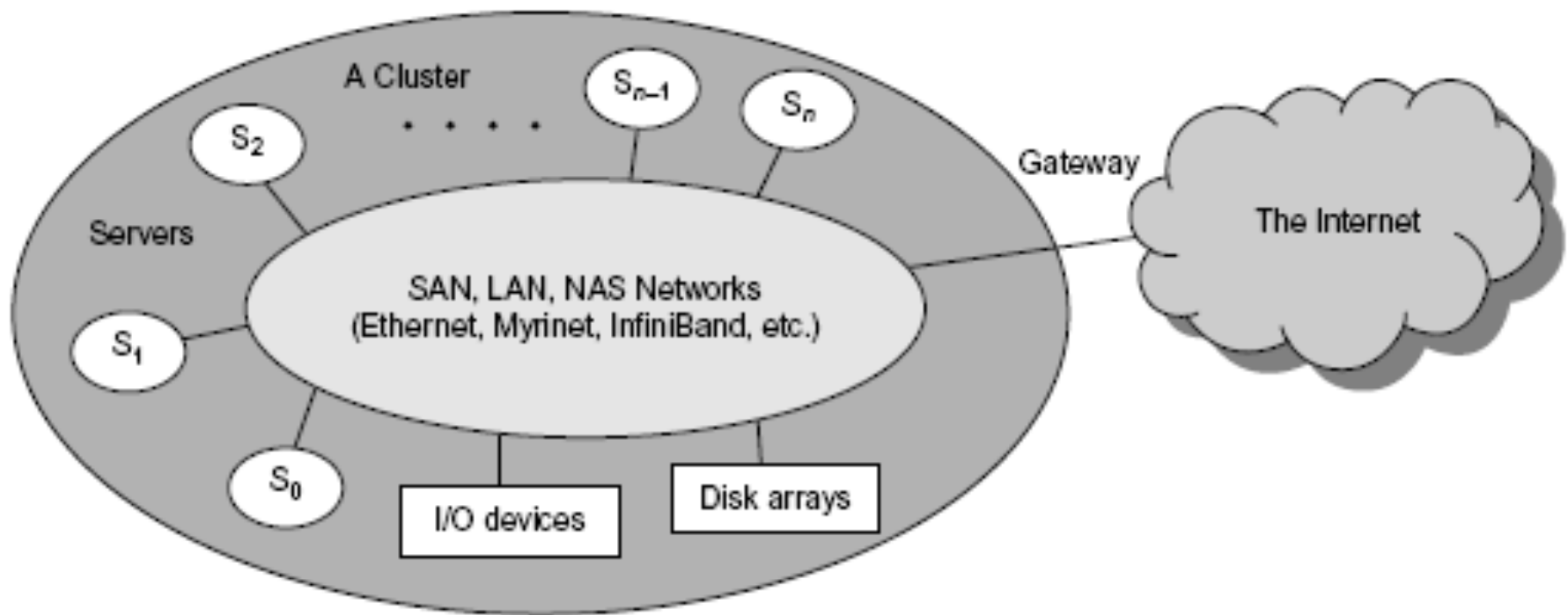
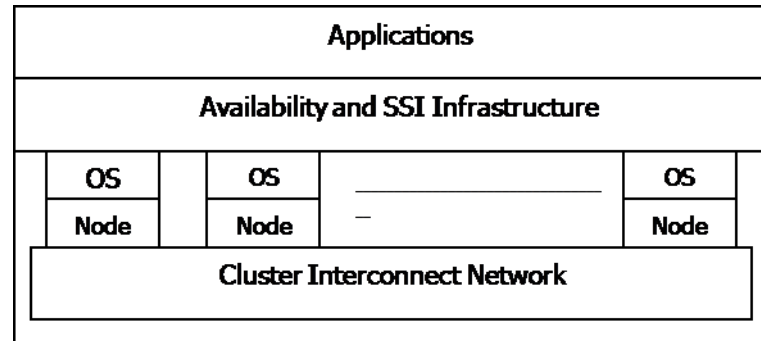


FIGURE 1.15

A cluster of servers interconnected by a high-bandwidth SAN or LAN with shared I/O devices and disk arrays; the cluster acts as a single computer attached to the Internet.

Cluster



- Multiple stand alone Machines Working in Parallel Connected by a High Speed LAN
- Single System Image (SSI)
- Centralized Job Management, Scheduling and the Homogeneous Resources Running 'Group Tasks'
- The Computing Nodes are a part of a single room (in the organization) with a very high degree of coupling
- The number of nodes much lesser as compared to the DCS
- Objective to minimize the TAT

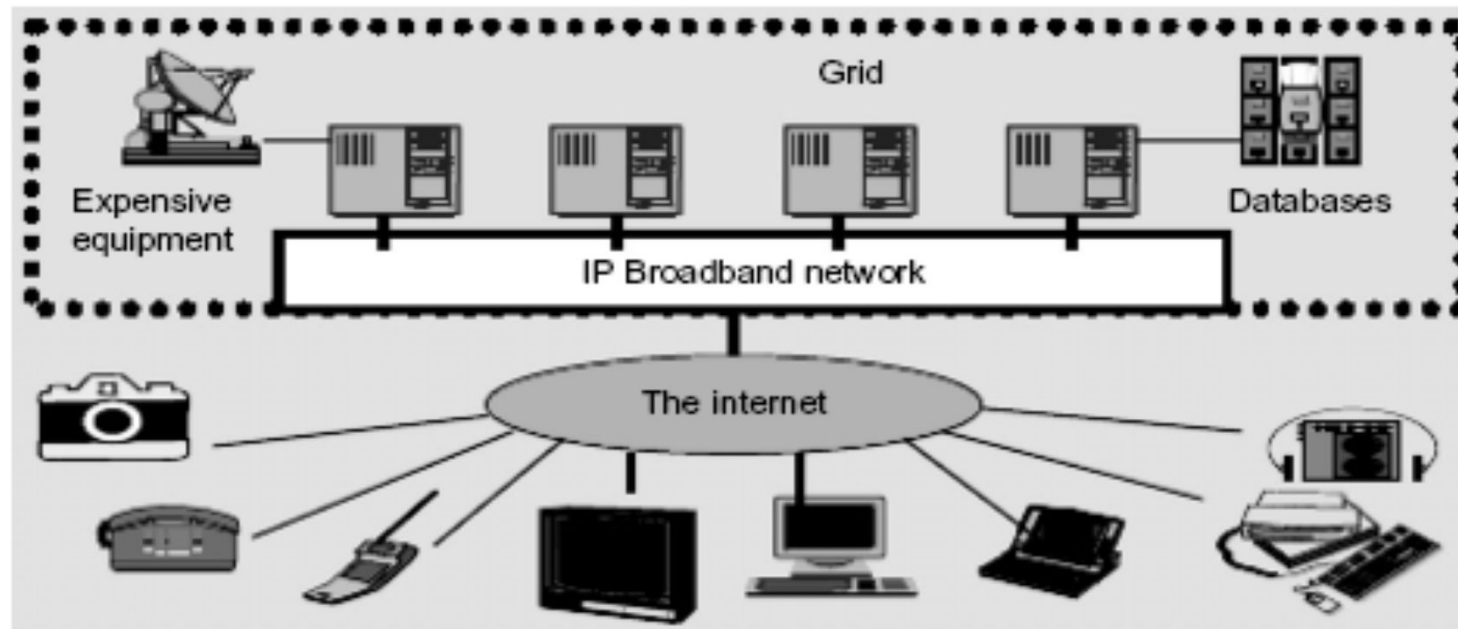


FIGURE 1.16

Computational grid or data grid providing computing utility, data and information services through resource sharing and cooperation among participating organizations.

System Users

Scientists and engineers using computation to accomplish Lab missions.

Intelligent Interface

A knowledge-based environment that offers users guidance on complex computing tasks.

Middleware

Software tools that enable interaction among users, applications and system resources.

Cluster Operating System

The software which coordinates the interplay of computers, networks and storage.

Supercomputing

Heterogeneous collection of high-performance computer hardware and software resources.

Networking

The hardware and software that permits communication among distributed users and computer resources.

Mass Storage

A collection of devices and software that allow temporary and long-term archival storage of information.

A Typical Computational Grid

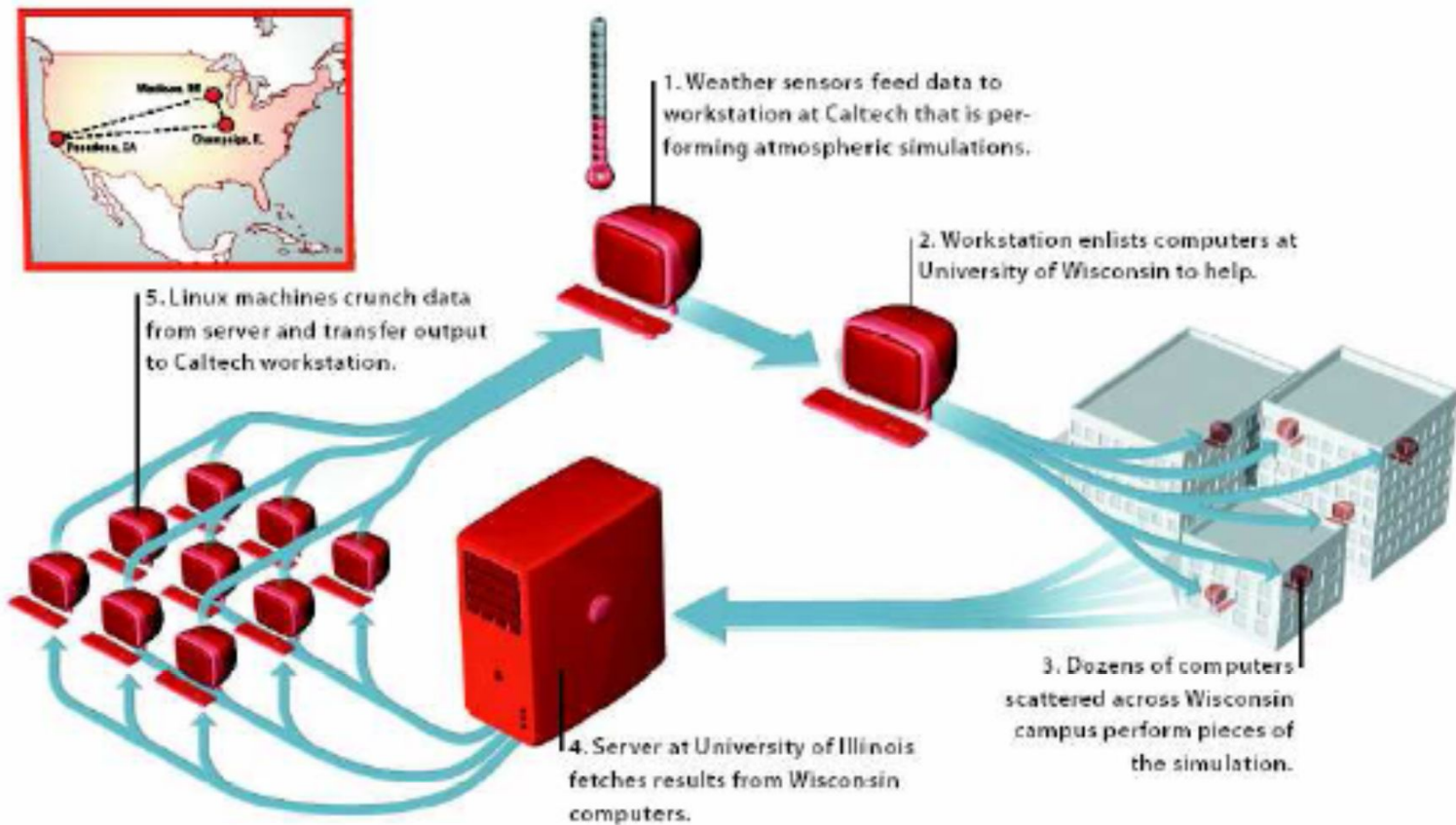


Figure 1.17 An example computational Grid built over specialized computers at three resource sites at Wisconsin, Caltech, and Illinois. (Courtesy of Michel Waldrop, "Grid Computing", IEEE Computer Magazine, 2000. [42])

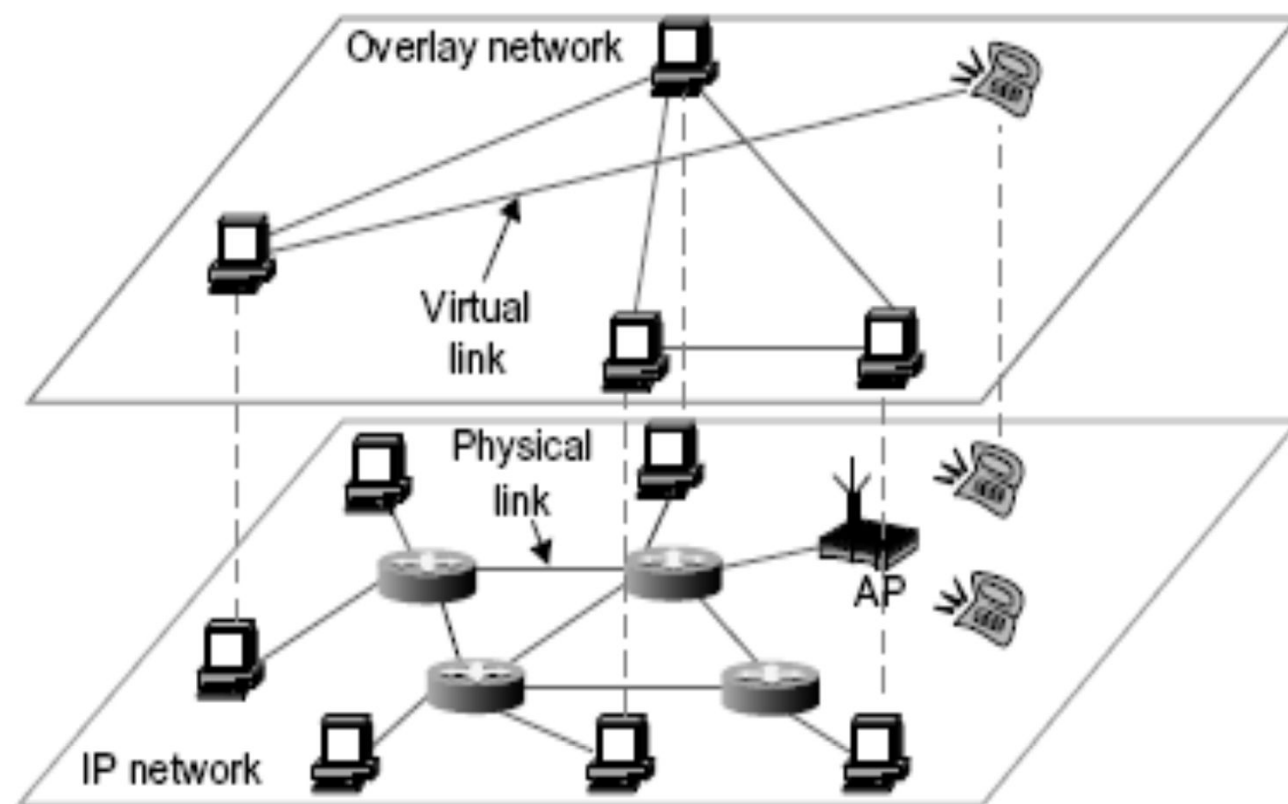


FIGURE 1.17

The structure of a P2P System by mapping a physical IP network to an overlay network built with virtual Links.

(Courtesy of Zhenyu Li, Institute of Computing Technology, Chinese Academy of Sciences, 2005)

Table 1.5 Major Categories of P2P Network Families [42]

System Features	Distributed File Sharing	Collaborative Platform	Distributed P2P Computing	P2P Platform
Attractive Applications	Content distribution of MP3 music, video, open software, etc.	Instant messaging, collaborative design and gaming	Scientific exploration and social networking	Open networks for public resources
Operational Problems	Loose security and serious online copyright violations	Lack of trust, disturbed by spam, privacy, and peer collusion	Security holes, selfish partners, and peer collusion	Lack of standards or protection protocols
Example Systems	Gnutella, Napster, eMule, BitTorrent, Aimster, KaZaA, etc.	ICQ, AIM, Groove, Magi, Multiplayer Games, Skype, etc.	SETI@home, Geonome@home, etc.	JXTA, .NET, FightingAid@home, etc.

The Cloud

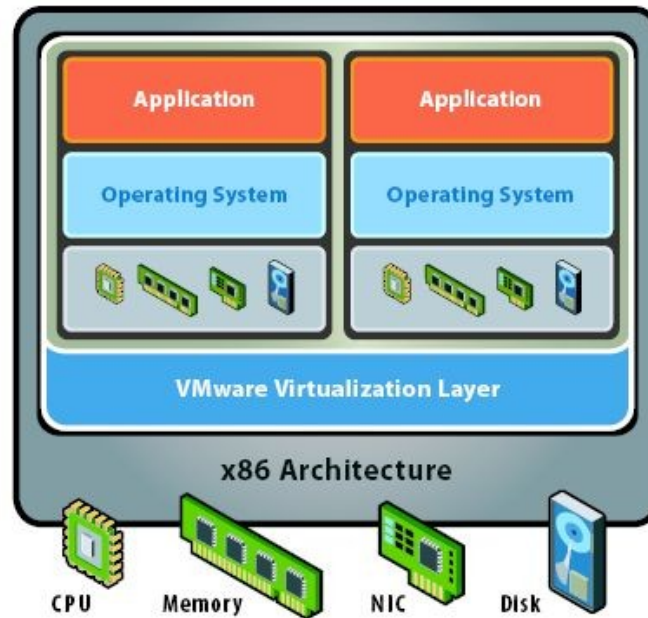
- Historical roots in today's Internet apps
 - Search, email, social networks
 - File storage (Live Mesh, Mobile Me, Flickr,...)
- A cloud infrastructure provides a framework to manage scalable, reliable, on-demand access to applications
- A cloud is the “invisible” backend to many of our mobile applications
- A model of computation and data storage based on “pay as you go” access to “unlimited” remote data center capabilities



Virtual Machines and Virtualization

- Conventional computer has single OS image
- This results in tightly coupled application software to a specific hardware platform
- Virtual machines offer novel solutions to utilize underutilized resources, application inflexibility, software manageability and security concerns in existing physical machines
- Host machine has the physical hardware
- Between virtual machine and host platform lies the middleware called Virtual Machine Monitor (VMM)
- Native VM with VMM in privileged mode known as Hypervisor

Virtual Machine Architecture



After Virtualization:

- Hardware-independence of operating system and applications
- Virtual machines can be provisioned to any system
- Can manage OS and application as a single unit by encapsulating them into virtual machines

(Courtesy of VMWare, 2010)

Primitive Operations in Virtual Machines:

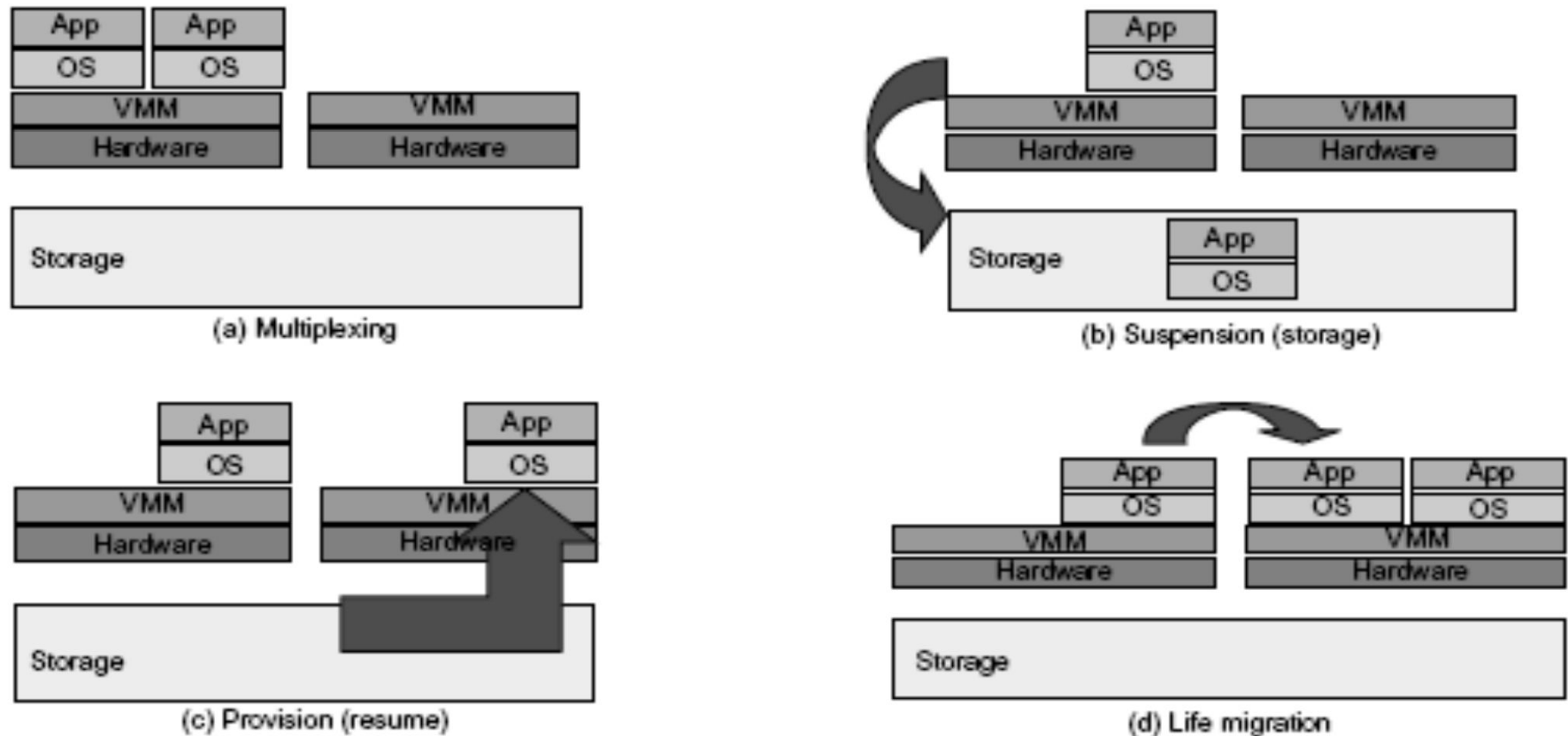


FIGURE 1.13

VM multiplexing, suspension, provision, and migration in a distributed computing environment.

(Courtesy of M. Rosenblum, Keynote address, ACM ASPLOS 2006 [41])

Concept of Virtual Clusters

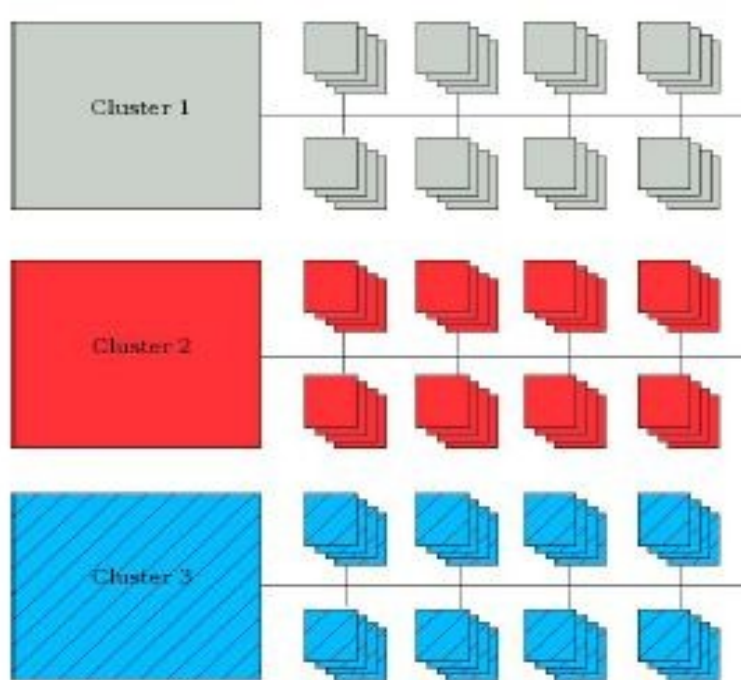


Fig. 1. A Campus Area Grid

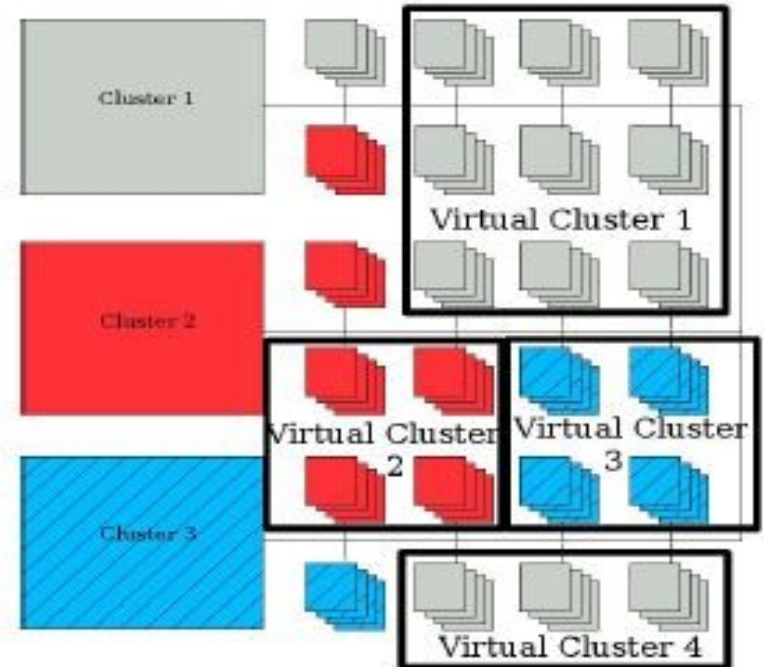
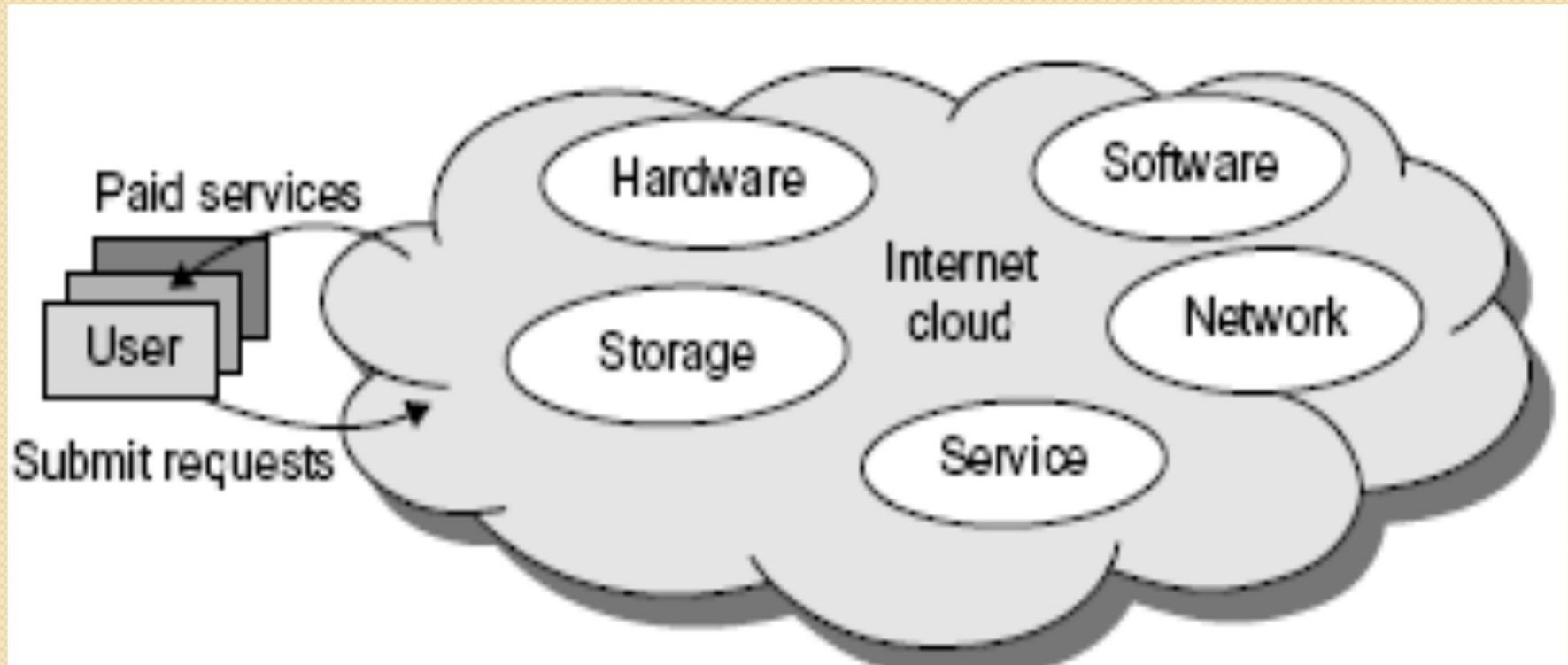


Fig. 2. Virtual machines in a cluster environment

(Source: W. Emeneker, et al, "Dynamic Virtual Clustering with Xen and Moab, ISPA 2006, Springer-Verlag LNCS 4331, 2006, pp. 440-451)

Basic Concept of Internet Clouds



The Next Revolution in IT : Cloud Computing

Every 18 months?

- **Classical Computing**

- Hardware, System Software, Applications often to meet peak needs.
- Install, Configure, Test, Verify, Evaluate
- Manage
- ..
- Finally, use it
- \$\$\$\$....\$(High CapEx)

- **Cloud Computing**

- Subscribe
- Use



- \$ - pay for what you use, based on QoS

(Courtesy of Raj Buyya, 2012)

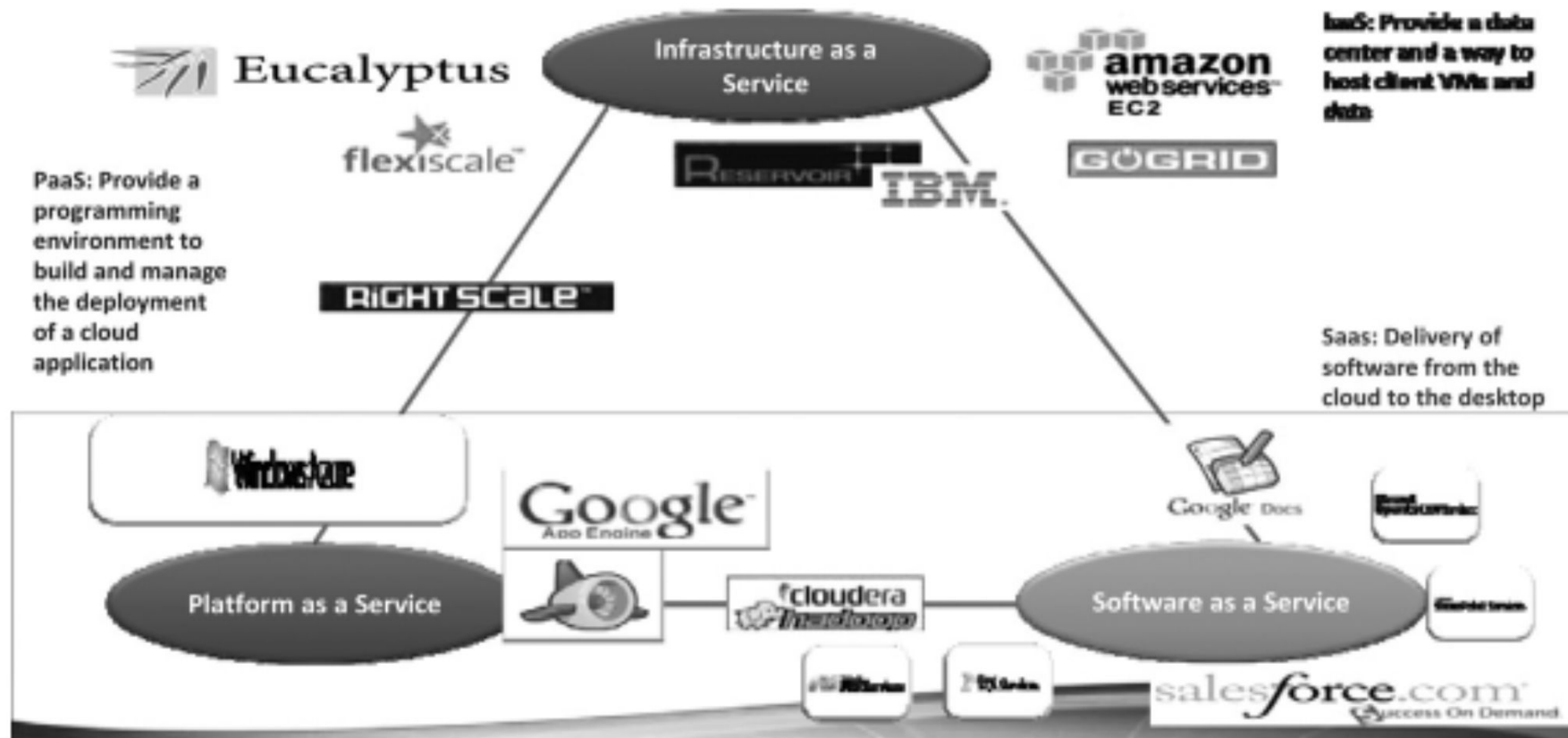


FIGURE 1.19

Three cloud service models in a cloud landscape of major providers.

(Courtesy of Dennis Gannon, keynote address at Cloudcom2010 [19])

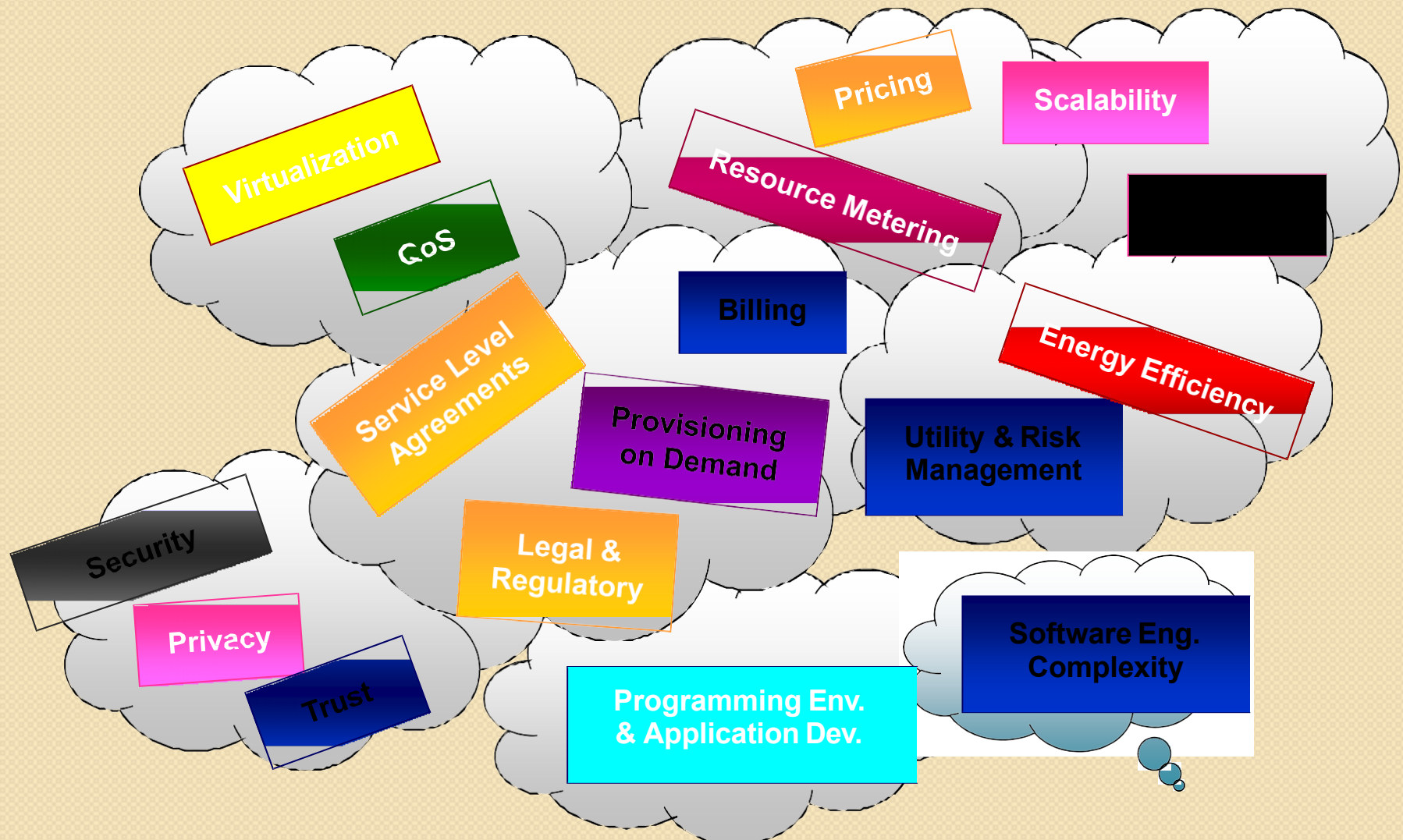
Cloud

Applications
Deployment Platform (Middleware)
Unified Resource View

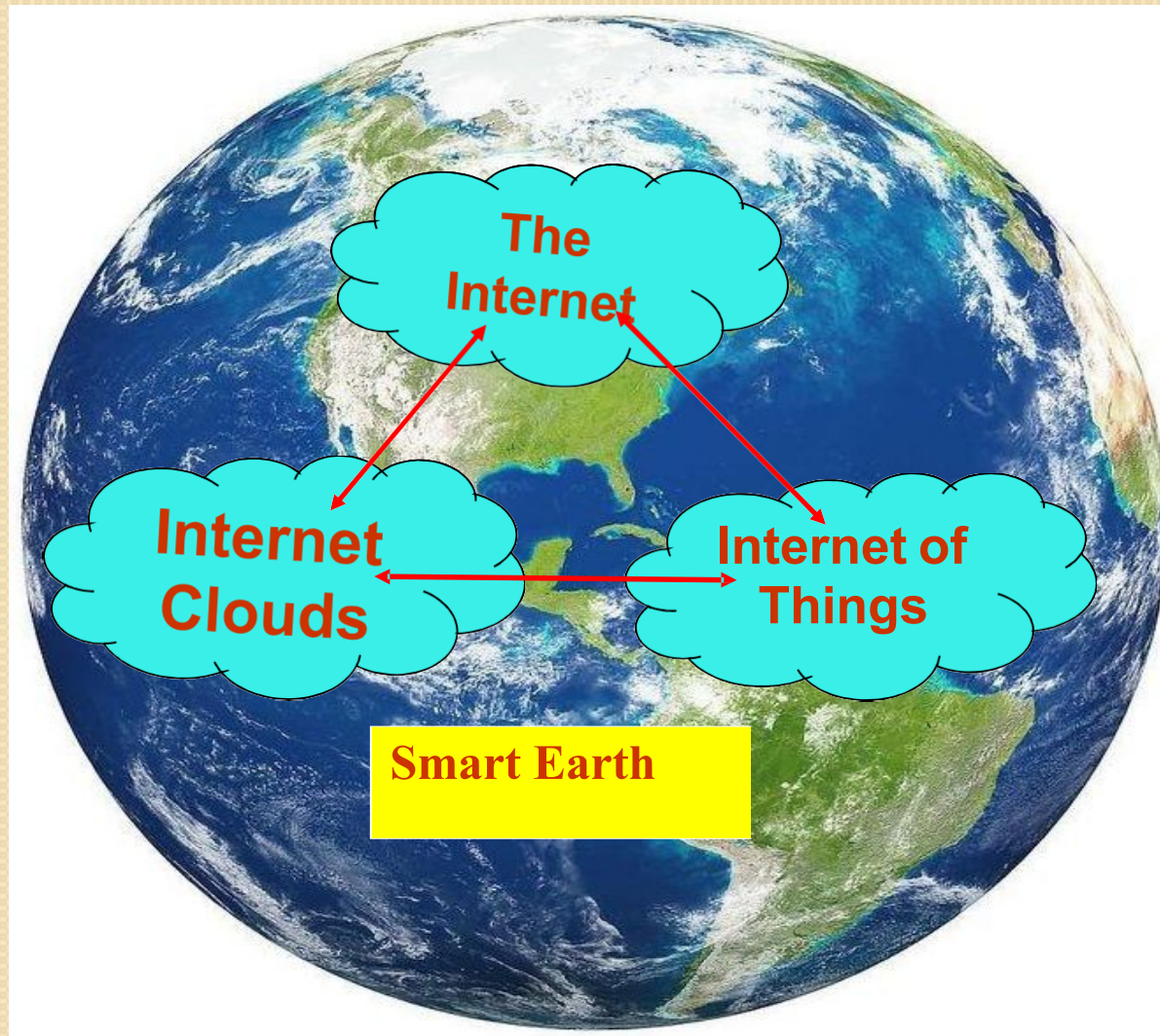
- **Organizations Participates on the Web (Cloud) to Provide Specialized Services to the Users**
- **User is Well Aware of his Requirements and Catering Organizations thereby Choosing the Services as Per his Willingness**
- **Eliminates the Cost of Purchasing and Managing Hardware and Software Needed to Design and Deploy Applications**
- **Helpful to Those who cannot Afford the Infrastructure Setup Cost on their Own for such kind of Services or the Expenditure is Enormous as Compared to the Returns Generated due to the Frequency of Use of Such Services**
- **Helpful for the Small and Medium Sized Companies while Paying only for what they use**

Cloud Computing Challenges:

Dealing with too many issues (Courtesy of R. Buyya)



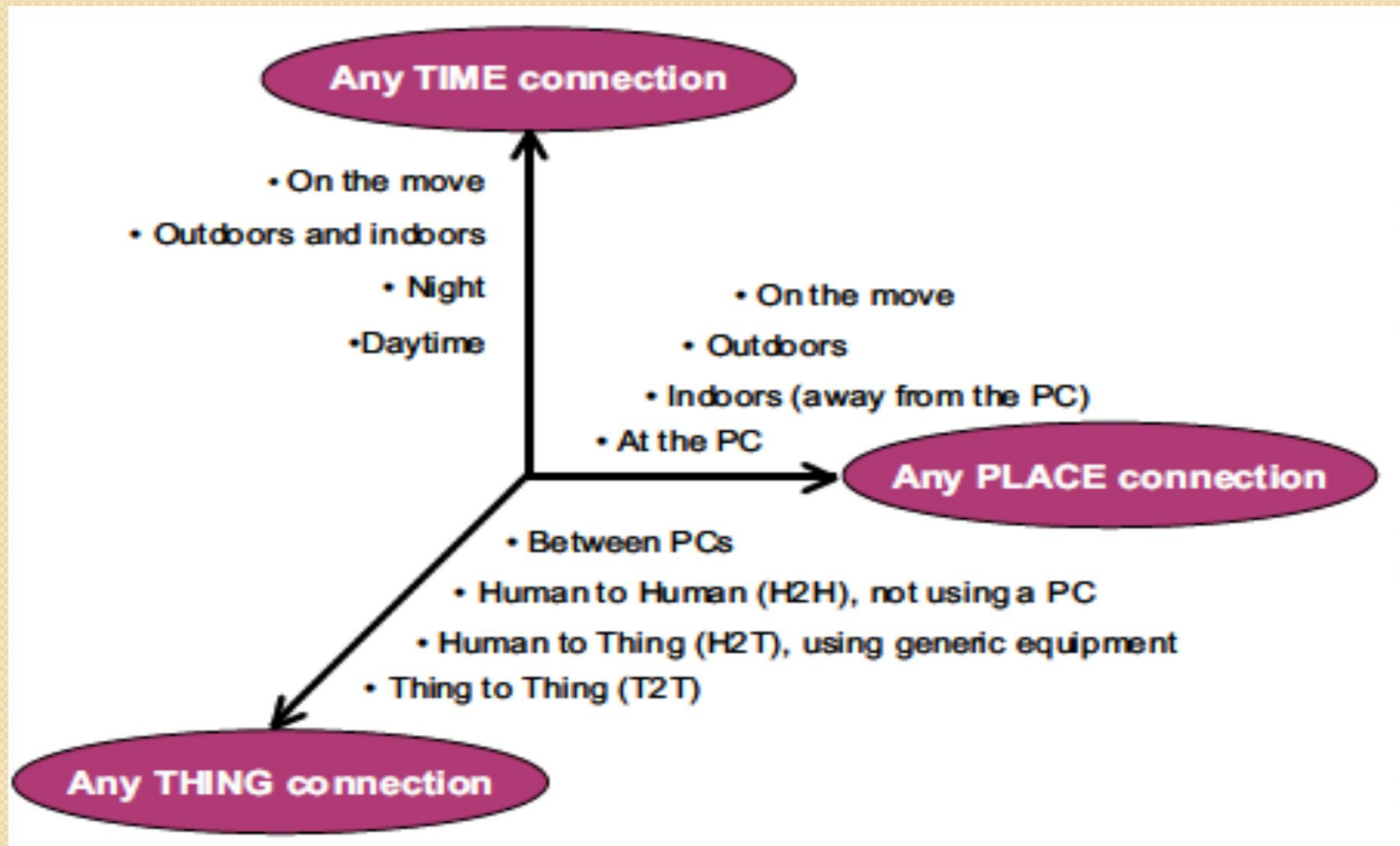
The Internet of Things (IoT)



**Smart
Earth:**

**An
IBM
Dream**

Opportunities of IoT in 3 Dimensions



(courtesy of Wikipedia, 2010)

System Scalability vs. OS Multiplicity

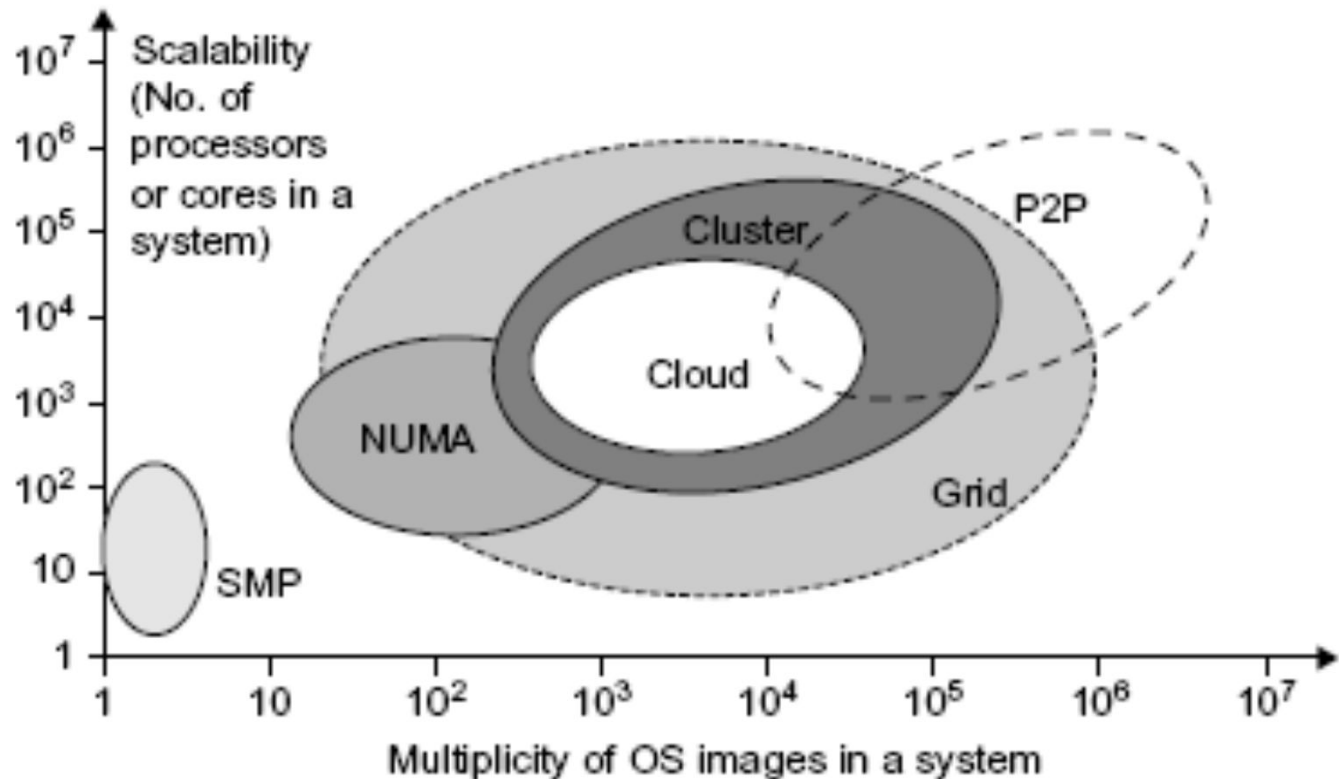


FIGURE 1.23

System scalability versus multiplicity of OS images based on 2010 technology.

Transparent Cloud Computing Environment

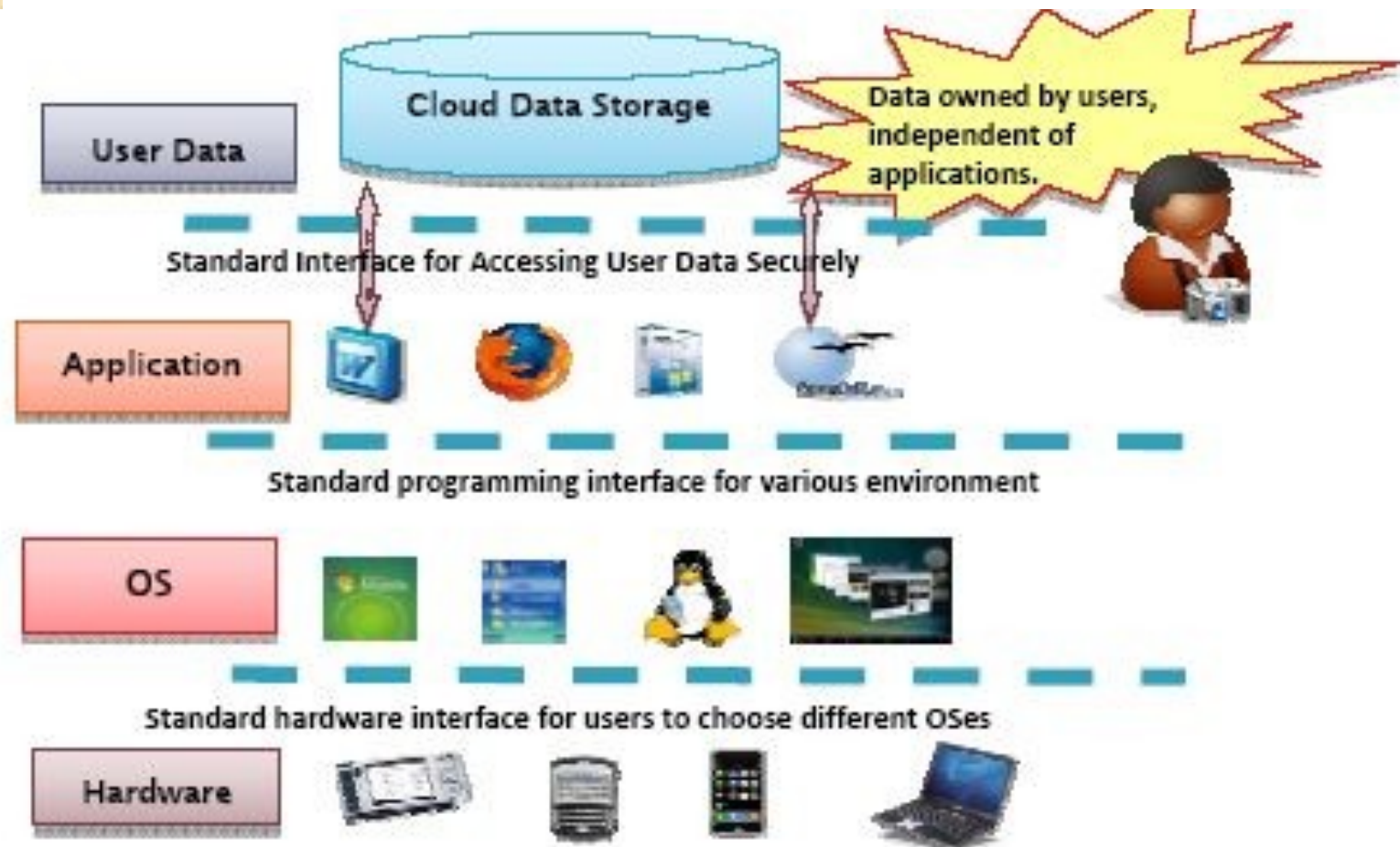


Figure 3 Transparent computing that separates the user data, application, OS, and hardware in time and space – an ideal model for future Cloud platform construction

Performance Metrics

- ✓ CPU Speed
- ✓ Dimensions of Scalability
 - ✓ Number of machines
 - ✓ Software scalability
 - ✓ Application scalability
 - ✓ Technology scalability
- ✓ Amdahl's Law and Gustafson's Law